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RESEARCH ARTICLE

Examining the Mediating Effects of Game User Experience Satisfaction on Continuance Intention: A Case Study of League of Legends

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ABSTRACT This research critically examines the relationship between game user experience satisfaction and continuance intention, focusing on dividing game user experience satisfaction into design and psychological factors and examining their mediating roles. Data were collected from 524 players of *League of Legends* and *League of Legends: Wild Rift* via online surveys. Using Partial Least Squares Structural Equation Modeling (PLS-SEM) and Multi-Group Analysis (PLS-MGA), the study examines factors such as usability/playability, narratives, play engrossment, enjoyment, creative freedom, audio aesthetics, personal gratification, social connectivity, visual aesthetics, and continuance intention. The findings demonstrate that desktop version players are more likely to have higher educational attainment and more extended gaming experience than mobile version players, with a partial mediating relationship between game user experience satisfaction and continuance intention. A significant observation is the differences between casual and core players in user experience satisfaction. In contrast, audio aesthetics have a more pronounced effect on enjoyment for casual players. Additionally, for core players, visual aesthetics have a more substantial impact on continuance intention, mediated by enjoyment, compared to casual players. However, no significant differences were observed across hypothesized paths based on gaming platforms. These insights underscore the importance of game user experience satisfaction in the mediating influence on players' continuance intention. Key design elements, such as enhancing audio and visual aesthetics, are identified as practical strategies to improve player engagement and loyalty, providing valuable insights for developers optimizing gaming experiences across platforms.

INDEX TERMS eSports, video game, user experience, continuance intention, League of Legends.

I. INTRODUCTION

In the wave of the digital age, video games and mobile devices have become essential components of people's lives, with significant differences in the frequency and duration of media tool usage compared to the past [1]. Video games not only integrate elements of entertainment and mass media but also give rise to what is referred to as the "Gamer Generation," a demographic that considers video games a part of daily life [2]. In this era, video games are viewed as a facet of human culture, profoundly influencing individuals' self-concepts, worldviews, social connections, and life goals [3], [4]. Historically, individuals could only experience the enjoyment of video games through devices such as personal computers or home gaming consoles. However, with the rapid proliferation of smartphones in the consumer electronics market, people can now play various video games anytime and anywhere through these mobile devices, leading to an expanding influence of video games on the industry [5], [6].

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The influence of the times extends beyond media and culture, as the rise of video games and the widespread adoption of the internet and social networking services have enhanced the economic and social impact of electronic sports (eSports) [7]. Multiplayer Online Battle Arena (MOBA) games within eSports have attracted a large fan base and significantly influenced contemporary game designers, with *League of Legends*, released in 2009 by *Riot Games*, becoming a focal point in the eSports scene [8]. As the number of players and viewers has increased, the audience for *League of Legends* live streams surpassed that of the National Basketball Association (NBA) in 2014. Furthermore, eSports was included as an event in the 2022 Asian Games, featuring *League of Legends*, marking the transformation of eSports from a niche interest to a global mainstream phenomenon and even an emerging industry that transcends the boundaries of traditional sports [9]. The real-time, high-frequency interactions in eSports have become a significant source of engagement and excitement for individuals seeking interactive and stimulating experiences in video games.

League of Legends is a highly popular MOBA game available on desktop platforms, as illustrated in FIGURE 1. The core gameplay involves two teams, each composed of five players, engaging in real-time combat, with each match typically lasting between 30 to 50 minutes [10]. The game offers various modes, including regular matches, ranked matches, and casual modes, where players select heroes with unique abilities and perform specific roles within their teams [11]. Success in the game relies on effective communication and strategic cooperation, with the ultimate objective being the destruction of the opposing team's base. Since randomly matched strangers often form teams, players must overcome communication and coordination challenges to achieve victory [12].



FIGURE 1. Screenshot of the desktop version of league of legends.

League of Legends: Wild Rift, released by Riot Games in 2020, is a mobile adaptation of the popular desktop game *League of Legends*, sharing the same setting and core mechanics as depicted in FIGURE 2. Although both games belong to the MOBA genre and follow identical fundamental rules, *League of Legends: Wild Rift* is specifically designed and optimized for mobile platforms. Unlike the desktop version, which relies on keyboard and mouse controls, it employs a more flexible and intuitive touch-based control system, incorporating features such as target locking, action buttons, and camera control for enhanced gameplay. Additionally, the game emphasizes a faster-paced experience with more frequent team battles, allowing players to engage in diverse challenges and enjoyable gameplay within relatively shorter match durations [13].



FIGURE 2. Screenshot of the mobile version of League of Legends: Wild Rift.

From the perspective of the video game industry, improving game design, enhancing the gaming experience, and segmenting players based on gaming data is are essential for evaluating player loyalty [14]. Studies of *League of Legends*, which has a large fan base, have shown that enhancing players' emotions and perceptions during gameplay can improve the quality of interaction between players and the game, serving as a basis for refining game design [15], [16]. Research on continuance intention in games commonly employs the Technology Acceptance Model (TAM) combined with Partial Least Squares Structural Equation Modeling (PLS-SEM) analysis, revealing that factors such as usability, perceived enjoyment, and immersion positively influence continuance intention [17], [18], [19]. Additionally, studies related to gamification have confirmed that games can influence players' behavioral intentions through enjoyment, creative thinking, and a sense of dominance [20], [21], [22]. While these studies provide a theoretical framework for examining gaming experiences, most of them have not thoroughly explored how specific game mechanics and aesthetics influence player experiences. Game mechanics, as the foundation of interaction, directly impact players' enjoyment and playability, which is particularly important when describing engaging experiences. On the other hand, game aesthetics—including visual, auditory, and narrative elements—significantly affect players' emotions, immersion, and sense of meaning, which are crucial for creating more affluent and more prosperous meaningful gaming experiences [23], [24]. Previous studies have primarily focused on higher-level relationships between variables such as motivation, satisfaction, and continuance intention, with limited attention given to how detailed game design elements act as mediators or directly

affect these variables [25], [26]. However, research has found that game elements like immersion and creativity can significantly enhance players' psychological need satisfaction, thereby increasing game satisfaction and continuance intention. Moreover, conceptualizing the gaming experience into distinct constructs of ''game design'' and ''user experience'' helps clarify the hierarchical relationship between design factors and psychological factors. Game design and systems merely provide the potential conditions for generating gaming experiences, but whether they transform into psychological experiences depends on users' voluntary acceptance and engagement [27], [28]. These findings emphasize that design factors and psychological factors operate at different levels and have distinct impacts on players' satisfaction and behavioral intentions. Such distinctions are valuable for establishing more structured and explanatory models, thereby offering a more comprehensive understanding of how gaming experiences influence players' continuance intentions.

In the digital age, differences in video games extend beyond the games themselves, with significant variations in user satisfaction regarding graphics and controls between desktop computers and smartphones [29], [30]. Researchers have found that a game's usability is a key factor influencing its appeal and that graphics, control differences, and gaming experience are potential factors that shape these perceptions [31]. When examining gaming experiences, the gaming platform itself is an essential factor that cannot be overlooked. For instance, in terms of system usability, studies suggest that the overall System Usability Scale scores for mobile platforms may be slightly higher than those for desktop platforms. However, a deeper analysis reveals that desktop users generally perceive the system as having better functionality integration and lower complexity. In contrast, mobile users are more likely to feel that additional technical support is needed during operation [32]. In evaluating playability, game designers must carefully consider cross-platform design differences to ensure that user experiences meet player expectations across various platforms. Desktop platforms tend to receive higher ratings for usability-related features, such as audiovisual presentation, game world visibility, screen layout, user interface responsibilities, and navigation design. This indicates that different platform interfaces and user habits can significantly influence players' perceptions of game fluidity and convenience [33], [34]. Researchers have classified players based on cognitive abilities and gaming time, finding differences in real-time responses, task handling, and gaming skills among different types of players [35], [36]. Compared to casual players, who often regard mobile games as tools for passing the time, core players place more value on the challenges and immersion provided by games, and their level of immersion in different gaming scenarios may differ from that of casual players [37]. Due to their more extended gaming experience and familiarity with the game world, core players are more likely to achieve higher levels of immersion and positive emotions. They may be more willing to engage deeply with games. In contrast, casual players may place greater importance on usability and immediate entertainment [38]. For both core and casual players, the quality and refinement of a game significantly affect game reputation and loyalty [14]. Thus, improving game design to enhance the gaming experience requires consideration not only of the game itself but also of how various factors—such as graphic design, platforms, and gaming experience—affect players' attitudes and experiences.

Based on the aforementioned theoretical perspectives, this study adopts a user experience satisfaction framework to explore the complex relationships between game design elements, psychological factors, and continuance intention. Unlike previous studies that primarily focused on a single platform, this research compares user experiences between the desktop and mobile versions of *League of Legends*. It clearly distinguishes game design elements (e.g., usability, aesthetics, and mechanics) from psychological factors (e.g., enjoyment, immersion, and social connectivity), providing an in-depth analysis of how their interactions produce differentiated impacts across various platforms and player types. Previous studies have indicated that mobile players generally report higher satisfaction in terms of usability/playability, visual aesthetics, and overall continuance intention compared to desktop players, suggesting that platform differences significantly influence user experience [39]. However, limited attention has been paid to the mediating role of psychological factors in these relationships. Therefore, this study aims to clarify the underlying sources of platform differences through a more detailed analysis, offering practical and valuable design recommendations for game developers to enhance player engagement and continuance intention across different gaming environments.

II. RESEARCH FRAMEWORKS AND HYPOTHESIS DEVELOPMENT

A. THEORETICAL BACKGROUND

This study primarily adopts the Game User Experience Satisfaction and continuance intention model [25] to understand the factors influencing players' continuance intention. The Game User Experience Satisfaction is developed based on the Game User Experience Satisfaction Scale (GUESS) [26], which thoroughly examines nine dimensions related to game evaluation: usability/playability, Narratives, play engrossment, enjoyment, creative freedom, audio aesthetics, personal gratification, social connectivity, and visual aesthetics. Overall, the model has adequately explored the relationship between Game User Experience Satisfaction and continuance intention, making it suitable for this study.

Specifically, usability/playability refers to the clarity of game objectives and intuitive controls; narratives involve the attractiveness of storylines and emotional connection with characters; audio aesthetics encompass the richness of auditory elements enhancing the gaming experience; and visual aesthetics pertain to the game's visual attractiveness. Play

engrossment captures players' sustained involvement and attention; enjoyment denotes the pleasure experienced during gameplay; creative freedom assesses the degree of freedom in exploration and personal expression; personal gratification indicates the motivational fulfillment and sense of achievement from game challenges; and social connectivity refers to the ease of interacting socially with other players.

In research on promoting game continuance, studies have used Self-Determination Theory to develop models that categorize influential factors into three primary constructs: Game Design, Activity Satisfaction, and Behavioral Engagement [27]. These frameworks posit that design factors shape psychological states, subsequently influencing behavioral outcomes such as game continuance. Similarly, prior studies indicate that game design elements—comprising game mechanics, narratives, and sensory aesthetics—directly affect psychological experiences, including immersion, enjoyment, creative freedom, gratification, and social interactions, thereby influencing players' continuance intention [28], [40].

Based on this theoretical foundation, we categorize usability/playability, narratives, audio aesthetics, and visual aesthetics as design factors. At the same time, play engrossment, enjoyment, creative freedom, personal gratification, and social connectivity are identified as psychological factors. We propose that these design elements serve as foundational drivers, facilitating positive psychological experiences, thereby directly influencing players' continuance intentions.

The hypothesized relationships in this study are presented in FIGURE 3.

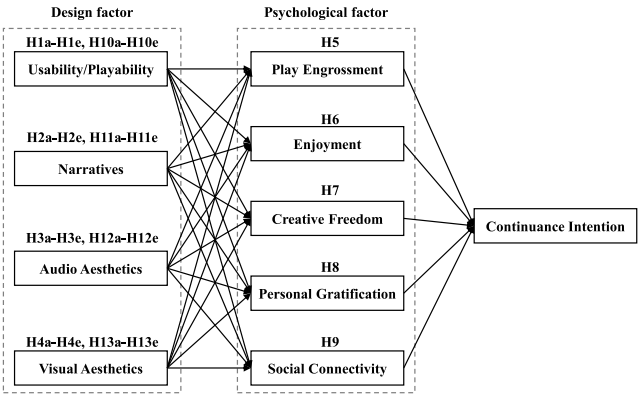


FIGURE 3. Research model.

B. HYPOTHESES' DEVELOPMENT

1) USABILITY/PLAYABILITY

Usability and playability significantly influence the overall user experience in game development [41]. In human-computer interaction, the primary driver of user satisfaction has been shown to be the system's operability, followed by the aesthetics of the interface [42].

Studies have demonstrated that the usability of video games enhances immersion, providing players with a more engaging experience [43]. Factors such as playability, narrative, and immersion are regarded as crucial elements that influence the gaming experience. Poor gameplay can not only detract from the overall experience but also prevent players from fully engaging with the story through interaction [44]. When video game usability and playability are overly complex, players may need to invest more effort in evaluation, which can diminish their sense of immersion in the game [45].

Research has shown that enjoyable gaming experiences in video games are driven by user-friendly interactive interfaces [46]. Functional experiences related to game controls and usability significantly impact the emotion of pleasure [47].

Gameplay is not only a key factor influencing the gaming experience but also indirectly affects players' sense of freedom within the game [48]. In particular, players experience the joy of freedom and control in the game world [49].

Studies using the Self-Determination Theory to examine the relationships between game motivation, gaming experience, and behavior have found significant differences in game experiences based on the psychological effects influenced by the game system [50]. Players tend to participate more willingly and perform better in games with high usability and playability, as opposed to games with frustrating controls and interface layouts [51].

In MOBA games, positive gaming experiences have been shown to enhance engagement in interactions with other players and are positively correlated with continuance intention [19]. The complexity of game content and controls has also been identified as a factor influencing players' attitudes toward social interactions within the game [52].

Research has consistently confirmed that usability and playability are crucial design factors in video games, positively impacting various aspects of the gaming experience. Based on this, we propose hypotheses H1a-H1e:

Hypothesis 1: Usability/playability has a positive impact on (a) play engrossment, (b) enjoyment, (c) creative freedom, (d) personal gratification, and (e) social connectivity.

2) NARRATIVES

Narratives are critical yet frequently overlooked design elements in video games. They profoundly influence players' emotional engagement and overall experience by intertwining seamlessly with other game components [53].

Game narratives provide players with unique emotional experiences, allowing them to project themselves and immerse deeply in the game world [54]. By constructing a vivid and realistic game world, narratives significantly impact player engagement [55]. Enhancing narrative features can spark players' curiosity, thereby increasing immersion [56].

The presentation of narratives in games fosters players' sense of involvement in the game world, evoking emotional responses, reflection, and an enhanced sense of well-being [57]. Well-crafted narratives enable players to deeply experience the game world deeply while also enjoying a pleasant emotional state [58]. Studies have confirmed that

narrative elements significantly influence players' enjoyment of the gaming experience [59].

Player creativity is viewed as an imaginative process, with game narratives and character backgrounds playing a crucial role in influencing players' emotional engagement [60]. The impact of narrative settings in games encourages deeper player reflection and stimulates limitless imagination about the story [61]. Providing deep narratives in games is also considered helpful for promoting player reflection on various topics [62].

The worldviews designed in video games offer players challenging scenarios, where they emerge as winners or losers in each game, thereby amplifying the psychological effects of overcoming challenges and showcasing self-expression [63]. Research has shown that players face different levels of challenges depending on the depth of the game's narrative [64]. Incorporating game narratives into learning environments can also elicit intrinsic empathy responses, leading to more excellent performance and motivation [65].

Game narratives are recognized as crucial design factors in video games, influencing players' imagination of the game world and fostering social interaction with other players [66]. Through the virtual world constructed by game narratives, players project their personalities and psychological states onto their in-game avatars, indirectly promoting social interaction with others [67]. Narrative design, in particular, can evoke players' emotions, leading to empathetic responses and encouraging positive social connections [68]. Video games also provide players with a virtual escape from the harsh realities of life, allowing them to engage in social interactions in ways different from the real world, free from the constraints of time and space, and alleviating feelings of loneliness [69].

Game narratives create an environment distinct from the real world. They not only immerse players in a virtual world, enhancing their sense of pleasure and imagination, but also influence self-perception and social interaction. Based on these factors, we propose hypotheses H2a–H2e:

Hypothesis 2: Narratives has a positive impact on (a) play engrossment, (b) enjoyment, (c) creative freedom, (d) personal gratification, and (e) social connectivity.

3) AUDIO AESTHETICS

Audio aesthetics—including music and sound effects—play an essential role beyond mere sensory stimulation, significantly shaping players' psychological states and emotional responses during gameplay [70]. Research has shown that using the right music and sound effects styles in appropriate situations can alter players' perceptions of the game [71].

In video game design, the role of audio in creating immersion is often overlooked compared to graphics. Yet, studies have confirmed that background music and sound effects can enhance the sense of immersion [72]. Background music tailored to the environment and context within video games significantly impacts players' immersion levels [73], [74]. audio aesthetics in games are considered to bring not only deeper and more meaningful immersion but also enjoyable emotional experiences [75].

Audio aesthetics can also increase positive emotions and reduce negative ones, leading to a more positive overall experience for players [76]. The game atmosphere is seen as a key element in the use of audio, where appropriate thematic congruence enhances players' physiological and psychological experiences [77]. Perceptual auditory feedback (PAF) in games helps create a situational atmosphere that brings players joy [78].

The atmosphere created by sound allows players to immerse themselves in the game's virtual world and imagine the environment in front of them, even extending to natural settings [79]. Sound is regarded as an underappreciated tool in providing an immersive environment in video games; it not only evokes pleasure but also complements visual content to shape players' perceptions and inspire imaginative experiences [80]. In stressful, anxious, or tense game scenarios, sound plays a crucial role in affecting players' psychological and physiological responses, fostering imaginative anticipation of uncertain or impending events [81].

Through experimental methods controlling audio conditions, studies have shown that adaptive audio can enhance players' sense of immersion and potentially influence their performance in maze navigation. Additionally, by comparing gameplay behaviors across different audio environments, research has explicitly demonstrated that audio types significantly affect players' gaming performance and decision-making processes. These findings suggest that audio design not only plays a critical role in enhancing immersion but also influences player behavior and interactive decision-making [74], [82]. Moreover, high-arousal music has been found to increase players' physiological arousal levels significantly and is associated with lower risk-taking behavior during the early stages of gameplay. This music style also affects players' sense of immersion and their perceptions of challenge and mastery. These results support the view that music can alter players' psychological states and impact their engagement and performance in specific gaming contexts, highlighting the essential role of audio design in shaping the gaming experience [83]. Furthermore, analyzing game interaction as a form of performative behavior highlights players' roles in co-creating the unique semiotic aesthetics of video game music, which directly contributes to enhancing immersion and enjoyment. Additionally, combining music with psychological regulation training has been found to influence individuals' physiological states effectively, indirectly supporting the positive impact of audio aesthetics on personal gratification [84], [85].

Analyzing virtual concerts within online games has shown that audio design not only enhances players' immersion and enjoyment but also fosters community co-creation between players and organizers. This co-creative environment further emphasizes the positive impact on creative freedom, personal gratification, and social connectivity [86]. The nature of multiplayer online games allows music to become a shared

experience, creating an environment where players' emotions synchronize and social connections are strengthened **[87].** Furthermore, compared to 2D audio, immersive 3D audio significantly enhances viewers' sense of social presence and parasocial interaction with the content, thereby increasing their enjoyment. This finding indirectly supports the influence of audio aesthetics on play engrossment, enjoyment, and social connectivity [88].

Sound in video games adds the finishing touch, crafting a compelling atmosphere that affects various aspects of the gaming experience. Based on this, we propose hypotheses H3a-H3e:

Hypothesis 3: Audio aesthetics has a positive impact on (a) play engrossment, (b) enjoyment, (c) creative freedom, (d) personal gratification, and (e) social connectivity.

4) VISUAL AESTHETICS

Visual aesthetics constitute a fundamental aspect of game design, directly influencing players' emotional responses, immersion, and social interactions, thereby enhancing the overall gaming experience [89].

The visual aesthetics of video games can evoke strong emotions in players, enhancing their sense of immersion [55]. Comparing user experiences in adventure game modes designed based on color psychology versus those designed purely by designers' intuition has shown that visuals crafted with psychological principles can significantly enhance players' gameplay immersion [90], [91]. Additionally, the visual appearance of virtual game characters, including facial expressions and animations, can promote players' situational empathy and immersion, which are critical factors influencing the overall user experience [92], [93]. Providing rich audiovisual feedback in games also helps to increase players' immersion and enjoyment [94].

Pleasant gaming experiences in video games are driven by aesthetically pleasing design content [46]. Enhancing the aesthetic design of games is considered one of the main drivers of gaming enjoyment [95]. Compared to static or monotonous feedback, ''juicy'' elements (rich dynamic visual feedback) in in-game visual design have been shown to deliver a more complete and enjoyable experience [96].

Game aesthetics in video games not only provide pleasure and emotional experiences but also extend players' imagination, creativity, and cognitive abilities [97]. The game world presents culture and space through visual aesthetics, allowing players to construct a virtual world in their minds that differs from reality [98]. Adding more fantastical visual aesthetic elements to games has been shown to stimulate players' imagination [99].

Visual aesthetics not only depict virtual world scenes but also allow players to project themselves onto game characters, using these avatars to escape reality and embody their desired or ideal selves [100]. The virtual avatar controlled by the player is considered an extension of the player, and the appearance and abilities of this avatar indirectly cause players to project themselves onto the virtual character, providing self-affirmation [101]. The emotions and colors associated with game characters have been proven to influence players' specific behaviors and emotional quality significantly [102].

In addition to playability, visual aesthetics are a central focus in video games. Engaging gameplay and a friendly visual atmosphere are believed to facilitate interaction among players [103]. Providing players with beautiful game environments not only makes the gaming process enjoyable but also creates a sense of realism and belonging in the virtual world, encouraging interaction and competition with others [104]. Visual aesthetics can foster familiarity through design and atmosphere, promoting prosocial interactions among players [105].

Visual aesthetics, composed of color and design shape players' first impressions of video games and are key factors influencing the gaming experience and atmosphere. Based on this, we propose hypotheses H4a-H4e.

Hypothesis 4: Visual aesthetics has a positive impact on (a) play engrossment, (b) enjoyment, (c) creative freedom, (d) personal gratification, and (e) social connectivity.

5) PLAY ENGROSSMENT

Play engrossment reflects players' depth of focus and sustained involvement during gameplay. It is a crucial psychological state influencing continued engagement, particularly prominent in immersive platforms like virtual reality [106].

Players' emotional involvement in MOBA games encourages more dedicated and sustained behavioral intentions to continue playing [107]. Various studies have referred to players' deep concentration on games as the ''flow experience,'' which has been confirmed as a primary factor influencing continuous gameplay [17]. Research has found that flow experience and perceived enjoyment significantly impact players' continuance intentions in massively multiplayer online games [18].

Play engrossment is considered a potential factor influencing the continuance intention to play the game. Therefore, we hypothesize:

Hypothesis 5: Play engrossment has a positive impact on continuance intention.

6) ENJOYMENT

Enjoyment—characterized by positive emotional experiences such as pleasure, emotional engagement, and arousal—is a pivotal factor driving players' willingness to engage continuously with MOBA games [107]. Pleasure emotions associated with enjoyment have also been shown to significantly influence word-of-mouth promotion about games [47].

In studies on MOBA games, motivational theories have been used to explore factors affecting user loyalty, with findings confirming that enjoyment has a significant impact [108]. Pleasant gaming experiences in video games have been demonstrated to influence continuance intention [46], [109]. Research has shown that players' gaming

experiences and satisfaction are positively correlated with continuance intention and gaming time [25], [110].

Positive emotions related to enjoyment have been confirmed by many studies to affect continuance intention for specific games. Therefore, we propose the following hypothesis:

Hypothesis 6: Enjoyment has a positive impact on continuance intention.

7) CREATIVE FREEDOM

Creative freedom in digital games offers players opportunities for imaginative expression and exploration, fostering a rich virtual environment filled with open-ended challenges and emotional engagement [60].

Research has shown that players' imaginative experiences triggered by game content influence their passion for the game [66]. Positive emotions associated with imagination have also been confirmed to positively affect the intention to replay in MMORPGs [111]. Two factors related to imaginative experiences—escapism and character projection—have been shown to impact continuance intention positively [107].

Video games not only provide players with a virtual world, but the creativity and imagination developed during gameplay also serve as emotional factors that encourage continued play. Therefore, we propose the following hypothesis:

Hypothesis 7: Creative freedom has a positive impact on continuance intention.

8) PERSONAL GRATIFICATION

Personal gratification, characterized by self-fulfillment and a sense of accomplishment, has been identified as a central motivational factor driving continued player engagement, particularly within mobile gaming contexts [112]. In eSports gameplay, long-term interactive engagement occurs, where personal gratification is not only a factor for players focusing on their performance but is also considered a self-reflective learning skill [113].

Competence satisfaction and autonomy satisfaction are regarded as psychological needs related to personal gratification, which can evoke positive emotions and motivate players to continue playing [27]. Various psychological factors associated with the individual, such as self-concept, self-congruity, and self-efficacy, have been confirmed to impact the intention to continue gaming [114]. Self-presentation, considered a form of utilitarian gratification where individuals express themselves to others, is also crucial for the continuance intention of eSports players [115].

Video games not only offer a world filled with entertainment and imagination but also provide a platform for self-expression and the achievement of self-gratification, which encourages players to continue playing. Based on this, we propose the following hypothesis:

Hypothesis 8: Personal gratification has a positive impact on continuance intention.

9) SOCIAL CONNECTIVITY

Social connectivity facilitated by video games uniquely enhances player engagement, differentiating gameplay experiences from other forms of digital entertainment by fostering interactive relationships among players [116].

Social interaction often facilitates self-expression in driving continuance intention [114]. This is further supported by identifying social motivation as a factor related to the continuance intention of online social games [117]. Social interaction in eSports games is a critical factor influencing the intention to continue playing [118].

Community awareness and relationship switching costs have been shown to influence the intention to continue online gaming [119]. In MOBA games, players' social interactions positively affect their willingness to continue playing [19].

These studies collectively indicate that social connections play a crucial role in shaping players' intentions to continue participating in eSports games. Based on this, we propose the following hypothesis:

Hypothesis 9: Social connectivity has a positive impact on continuance intention.

10) MEDIATING EFFECTS BETWEEN GAME USER EXPERIENCE SATISFACTION AND CONTINUANCE INTENTION

Usability and playability are not only significant factors influencing the gaming experience but have also been confirmed as key determinants affecting long-term game engagement [51]. Sensory experiences related to visual and auditory elements have been shown to influence behavioral intentions positively [107]. However, these studies lack an exploration of whether common design factors in video games are mediated by psychological factors that may arise in user experiences.

App design aesthetics, functionality, and quality are crucial for influencing user satisfaction, and these factors also indirectly affect continuance intention through satisfaction as a mediator [120]. Research on flow experience and perceived value in user engagement has found that a sense of control positively influences hedonic value, and user stickiness in eSports games is primarily influenced indirectly by hedonic value [121]. Positive functional experiences in online games have also been shown to positively impact enjoyment and a sense of dominance, which mediates word-of-mouth promotion [47]. Providing players with preferred worldviews and story backgrounds in games has been found to influence game loyalty indirectly through self-confidence and satisfaction [122]. Perceptual auditory feedback is identified as an essential design factor in video games, not only influencing immersion and enjoyment but also mediating continuance intention [78]. The graphical design of video games is also confirmed as a critical factor affecting the gaming experience, indirectly influencing continuance intention [123].

Design factors in video games not only impact the psychological aspects of the gaming experience but have also been confirmed by numerous studies to potentially

influence behavioral intentions through psychological mediators. Based on this, we propose the following hypotheses:

Hypothesis 10: The usability/playability on continuance intention mediates effect by (a) play engrossment (b) enjoyment, (c) creative freedom, (d) personal gratification, and (e) social connectivity.

Hypothesis 11: The narratives on continuance intention mediate effect by (a) play engrossment (b) enjoyment, (c) creative freedom, (d) personal gratification, and (e) social connectivity.

Hypothesis 12: The audio aesthetics on continuance intention mediates effect by (a) play engrossment (b) enjoyment, (c) creative freedom, (d) personal gratification, and (e) social connectivity.

Hypothesis 13: The visual aesthetics on continuance intention mediates effect by (a) play engrossment (b) enjoyment, (c) creative freedom, (d) personal gratification, and (e) social connectivity.

III. METHODS

A. PROCEDURE

Social media platforms are considered essential sources for understanding users [124]. Research on online multiplayer games in Taiwan has primarily recruited participants through social networking services like the PTT Bulletin Board System, Bahamut Game Forum, and Dcard [125], [126], [127].

Thus, this study conducted a convenience sampling survey from January 2, 2024, to January 31, 2024, by recruiting participants through frequently visited platforms, including the PTT Bulletin Board System, Bahamut Game Forum, and Dcard. To avoid differences caused by factors such as different game developers, game types, and story backgrounds, the study focused on players aged 20 and above who had played the same games developed by Riot Games, Inc., specifically *League of Legends* for the desktop platform and *League of Legends: Wild Rift* for the mobile platform. An online questionnaire was used to collect the data.

The online survey began with a cover page explaining the study's details, followed by a consent question on the next page, asking participants whether they agreed to continue with the questionnaire before proceeding.

B. SAMPLE

The target population for this study was residents of Taiwan, with a total population of 23,420,442 as of December 2023 [128]. Using the sample size calculator provided by Rao Soft Inc., we selected a 95% confidence level and a 5% margin of error, with a population size of 23,420,442, resulting in a recommended sample size of 385 [129]. Ultimately, we collected 524 valid questionnaires, and the average survey duration was 7 minutes, as shown in TABLE 1.

It is noteworthy that most study participants were male (74.4%), aged between 20 and 30 (73.5%), and primarily used the desktop version to play the game.

Additionally, to differentiate between different types of participants based on gaming experience and weekly playtime, this study classified players according to previous research. Participants with more than 4 years of gaming experience and less than seven hours of weekly playtime were defined as casual players, while those with more than 4 years of gaming experience and more than seven hours of weekly playtime were defined as core players [35], [36].

TABLE 1. Demographic characteristics of the sample. (N = 524).

Demographic variable	n (%)
Gender	
Male	393(74.4)
Female	135(25.6)
Age	
≥20 and <30 Years old	385(73.5)
≥30 and <40 Years old	129(24.6)
≥40 and <50 Years old	10(1.9)
Education	
High school or below	36(6.9)
University	383(73.1)
Graduate Schools	105(20)
Occupation	
Labor force	394(75.2)
Non-labor force	81(15.5)
Unemployed	49(9.4)
Annual income	
<NT$ 300,000	200(38.2)
NT$ 300,000 - 599,999	191(36.5)
≥ NT$ 600,000	133(25.4)
Gaming Experience	
<2 Years	75(14.3)
≥2 and <4 Years	115(21.9)
≥4 Years	334(63.7)
Weekly Gaming Hours	
<7 Hours	213(40.6)
≥7 and <14 Hours	192(36.6)
≥14 and <21 Hours	57(10.9)
≥21 Hours	62(11.8)

C. MEASURES

This study conducted a questionnaire survey covering four main sections: participant demographics, gaming experience, game user experience satisfaction, and continuance intention. The items measuring game user experience satisfaction and continuance intention are presented in TABLE 2.

The first section of the questionnaire included demographic variables such as gender, age, education level, and annual income. The second section covered game type, gaming experience, and frequency of play. The third section utilized the GUESS-18 of GUESS short version [130], which comprised 18 items across nine dimensions: usability/playability, narratives, play engrossment, enjoyment, creative freedom, audio aesthetics, personal gratification, social connectivity, and visual aesthetics. Each of the nine dimensions consists of two items, and the score for each dimension was calculated as the average of the two items. The fourth section assessed the continuance intention scale [25], using a scale with four items focused on a single dimension, with the overall score calculated as the average of these items.

TABLE 2. Evaluation questionnaire.

Constructs	Statements	Label
Usability/playability	I find the controls of the game to be straightforward.	U1
	I find the game's interface to be easy to navigate.	U2
Narratives	I am captivated by the game's story from the beginning.	N1
	I enjoy the fantasy or story provided by the game.	N2
Play engrossment	I feel detached from the outside world while playing the game.	PE1
	I do not care to check events that are happening in the real world during the game.	PE2
Enjoyment	I think the game is fun.	ENJ1
	I feel bored while playing the game. (REVERSE CODE)	ENJ2
Creative freedom	I feel the game allows me to be imaginative.	CF1
	I feel creative while playing the game.	CF2
Audio aesthetics	I enjoy the sound effects in the game.	AA1
	I feel the game's audio (e.g., sound effects, music) enhances my gaming experience.	AA2
Personal gratification	I am very focused on my own performance while playing the game.	PG1
	I want to do as well as possible during the game.	PG2
Social connectivity	I find the game supports social interaction (e.g., chat) between players.	SC1
	I like to play this game with other players.	SC2
Visual aesthetics	I enjoy the game's graphics.	VA1
	I think the game is visually appealing.	VA2
Continuance intention	In the future, I will continue to play the game.	CI1
	In the future, I will play the game often.	CI2
	I will say the advantages of the game to other people.	CI3
	I will recommend the game to other people.	CI4

Grounded in the theoretical framework that examines game motivation, satisfaction, and the continuance intention, the continuance intention scale was derived from previous studies to maintain consistency with established research [131].

Responses for the third and fourth sections were measured using a 7-point Likert scale, with each point anchored to a response label (e.g., 1 = strongly disagree, 4 = neutral, 7 = strongly agree).

The length of a questionnaire is considered an essential factor influencing response rate, attention, and response quality, with particular emphasis on the quality of the questionnaire content itself [132], [133], [134]. However, excessively short questionnaires, including those consisting of only a single item, may address issues related to survey length but can severely compromise reliability and validity [135], [136]. Studies have demonstrated that two- or three-item questionnaires can achieve similar psychometric performance compared to five-item scales [137]. The original 55-item GUESS (GUESS-55) requires approximately 10 to 15 minutes to complete, whereas the short-form version only takes 3 to 5 minutes, providing a concise format while maintaining satisfactory reliability and validity [26], [130]. Additionally, studies focusing on virtual reality have confirmed that the GUESS-18 version demonstrates acceptable reliability and validity [138].

D. STATISTICAL ANALYSIS

This study utilized the Chi-square test through SPSS to analyze the differences in demographic characteristics between desktop and mobile players. It examined whether the distribution of variables, including gender, age, education level, occupation, annual income, and gaming behavior, significantly differed across the two platform groups.

This study employed PLS-SEM for quantitative model analysis. The data analysis process was conducted using SmartPLS 4.1.1.1 [139].

This study employed Partial Least Squares Multi-Group Analysis (PLS-MGA) to examine the consistency of different paths. PLS-MGA is a non-parametric statistical method used to compare model path coefficients between different groups [140].

To ensure the reliability and validity of the questionnaire, we carefully considered the characteristics of the measurement scale. When only two items measure a construct, Cronbach's Alpha is mathematically equivalent to the Pearson correlation (r) between the two items, meaning that it may not fully capture the internal consistency of the construct and could either underestimate or overestimate the actual reliability. Therefore, this study adopts the Spearman-Brown formula, which is particularly suitable for two-item scales. By applying split-half reliability correction, this method provides a more robust estimation of reliability [141]. Additionally, this study applies Guttman split-half reliability, which is very similar to Cronbach's Alpha and is also used to estimate internal consistency and reliability [142], [143]. Unlike Cronbach's Alpha, Guttman split-half reliability does not assume tau-equivalence, making it a more reliable measure when item variances are unequal [144].

Through these complementary reliability assessments, this study ensures that the measurement model maintains adequate internal consistency and stability, even with two-item constructs.

E. ETHICS STATEMENT

The Human Research Ethics Committee of National Chung Cheng University approved this study on January 2, 2024 (Approval No. CCUREC112121802).

This study was conducted in accordance with the Declaration of Helsinki [145]. The invitation text and online questionnaire page provided a detailed description of the study's introduction, background, objectives, anonymity statement, and survey description. All participants gave informed consent, and the survey was conducted anonymously.

IV. RESULTS

A. DEMOGRAPHIC CHARACTERISTICS OF PARTICIPANTS

This study conducted a Chi-square test to analyze the demographic characteristics between desktop and mobile users, and the results are presented in TABLE 3.

The analysis revealed no significant differences in gender and age distribution between desktop and mobile players ($\chi^2 = 0.3$, $p = .557$; $\chi^2 = 1.2$, $p = .553$), with the majority of users aged 20-29, accounting for 72.4% of desktop players and 77.5% of mobile players. However, a significant difference was found in education level across different device types ($\chi^2 = 6.3$, $p = .043$). A higher proportion of mobile players had a high school education or below (10.8%) compared to desktop players (5.8%). In comparison, desktop players had a higher percentage of individuals with graduate-level education (21.8% vs. 13.5%). The distribution of occupation and annual income showed no significant differences between the two groups ($\chi^2 = 3.4$, $p = .179$; $\chi^2 = 1.7$, $p = .433$), indicating comparable distribution across labor markets and income levels. Furthermore, game experience differed significantly between the two groups ($\chi^2 = 137$, $p < .001$), with most desktop players having over four years of gaming experience (75.5%), In contrast, the majority of mobile players had two to four years of experience (59.5%). However, no significant difference was found in weekly gaming hours ($\chi^2 = 2.1$, $p = .550$), suggesting that the level of gaming engagement was generally similar between the two groups.

Overall, the demographic characteristics of desktop and mobile players are broadly similar, with significant differences observed only in education level and gaming experience. These findings suggest that mobile players are more likely to belong to groups with lower educational attainment and shorter gaming experience. In contrast, desktop players tend to have a more extended gaming history.

TABLE 3. Demographic comparisons between gaming platforms.

Demographic variables	Gaming Platforms., n (%) Desktop (n=413)	Mobile (n=111)	Chi-square test χ2 (df)	*p*-value
Gender			0.3 (1)	.557
Male	309(74.8)	80(72.1)		
Female	104(25.2)	31(27.9)		
Age			1.2 (2)	.553
≥20 and <30 Years old	299(72.4)	86(77.5)		
≥30 and <40 Years old	106(25.7)	23(20.7)		
≥40 and <50 Years old	8(1.9)	2(1.8)		
Education			6.3 (2)	.043
High school or below	24(5.8)	12(10.8)		
University	299(72.4)	84(75.7)		
Graduate Schools	90(21.8)	15(13.5)		
Occupation			3.4 (2)	.179
Labor force	310(75.1)	84(75.7)		
Non-labor force	60(14.5)	21(18.9)		
Unemployed	43(10.4)	6(5.4)		
Annual income			1.7 (2)	.433
<NT$ 300,000	154(37.3)	46(41.4)		
NT$ 300,000 - 599,999	149(36.1)	42(37.8)		
≥ NT$ 600,000	110(26.6)	23(20.7)		
Gaming Experience			137 (2)	<.001
<2 Years	52(12.6)	23(20.7)		
≥2 and <4 Years	49(11.9)	66(59.5)		
≥4 Years	312(75.5)	22(19.8)		
Weekly Gaming Hours			2.1 (3)	.550
<7 Hours	173(41.9)	40(36)		
≥7 and <14 Hours	145(35.1)	47(42.3)		
≥14 and <21 Hours	46(11.1)	11(9.9)		
≥21 Hours	49(11.9)	13(11.7)		

χ2 = chi-square test statistic, df = degrees of freedom.

B. MEASUREMENT MODEL EVALUATION

This study used factor loadings, Cronbach's alpha, Spearman-Brown, composite reliability (CR), and Average Variance Extracted (AVE) to assess reliability and validity, as shown in TABLE 4.

The factor loadings for all items ranged from 0.782 to 0.946. Factor loadings above 0.5 are considered practically significant, and values above 0.7 are regarded as having a clear structure and objective. A factor loading of 0.6 or above is recommended for studies with a sample size greater than 85 [146].

Cronbach's alpha (α), Spearman-Brown coefficient (SB), and Guttman Split-Half Reliability (Gut.) were used to detect the internal consistency of the research sample. Except for PE ($\alpha = .597$, SB = .597, Gut. = .596), ENJ ($\alpha = .640$, SB = .640, Gut. = .622), and SC ($\alpha = .545$, SB = .545, Gut. = .528), all other constructs exhibited reliability coefficients exceeding 0.7, indicating acceptable internal consistency. In most studies, a coefficient above 0.70 is considered to indicate sufficient reliability and internal consistency. At the same time, values from 0.5 to 0.7 are regarded as moderate reliability, and values below 0.5 are considered low reliability [147], [148].

All CR values ranged from 0.814 to 0.941, and all AVE values ranged from 0.656 to 0.889. Composite reliability is recommended to be above 0.6, and the Average Variance Extracted (AVE) for each construct should exceed the minimum standard of 0.50 [149].

According to the principles of the Fornell-Larcker criterion, the Average Variance Extracted (AVE) for a construct should be higher than the squared correlation values between that construct and other constructs. This indicates that the construct explains the variance of its indicators more effectively than it correlates with other constructs [149].

As shown in TABLE 5, the AVE for each construct and its corresponding indicators (ranging from 0.810 to 0.943) exceeded the squared correlation values between the construct and other constructs (ranging from 0.091 to 0.607). This observation suggests that each construct demonstrated good discriminant validity among its indicators.

The Heterotrait-Monotrait Ratio of Correlations (HTMT) test was used to assess the discriminant validity of constructs in the structural equation model. By comparing the internal correlations within constructs and correlations between constructs, the test determines whether they are measured distinctly. Generally, HTMT values should be less than 1, with more conservative thresholds of 0.90 or 0.85 indicating sufficient differences between constructs, allowing them to be accurately distinguished [150], [151].

TABLE 4. Factor loading and reliability.

Variables	Items	FL	α	SB	Gut.	CR	AVE
U/P	U1	.868	.757	.757	.749	.890	.802
	U2	.922					
N	N1	.930	.844	.844	.844	.928	.865
	N2	.930					
AA	AA1	.937	.864	.864	.862	.936	.880
	AA2	.940					
VA	VA1	.946	.875	.875	.874	.941	.889
	VA2	.940					
PE	PE1	.892	.597	.597	.596	.829	.709
	PE2	.788					
ENJ	ENJ1	.918	.640	.640	.622	.841	.727
	ENJ2	.782					
CF	CF1	.941	.867	.867	.867	.937	.882
	CF2	.938					
PG	PG1	.932	.836	.836	.831	.924	.859
	PG2	.921					
SC	SC1	.804	.545	.545	.528	.814	.686
	SC2	.853					
CI	CI1	.784	.825	-	-	.884	.656
	CI2	.809					
	CI3	.819					
	CI4	.828					

U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention, FL = Factor Loadings, α = Cronbach's Alpha, SB = Spearman-Brown, Gut. = Guttman Split-Half Reliability, CR = Composite Reliability, AVE = Average Variance Extracted.

TABLE 5. Discriminant validity - Fornell & Larcker.

Constructs	U/P	N	AA	VA	PE	ENJ	CF	PG	SC	CI
U/P	.896									
N	.229	.930								
AA	.267	.365	.938							
VA	.300	.382	.527	.943						
PE	.133	.243	.130	.128	.842					
ENJ	.234	.254	.412	.509	.112	.853				
CF	.230	.484	.400	.440	.209	.427	.939			
PG	.125	.091	.280	.347	.167	.383	.221	.927		
SC	.328	.200	.283	.405	.142	.391	.339	.241	.829	
CI	.310	.359	.462	.610	.185	.607	.473	.343	.482	.810

U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.

As shown in TABLE 6, the highest similarity between all constructs was 0.813, meeting the recommended threshold and indicating adequate discriminant validity.

C. STRUCTURAL MODEL EVALUATION

In conducting the structural model evaluation, it is essential to examine whether common method bias (CMB) exists within the model. CMB is a potential issue that may artificially inflate correlations between measurement variables, thereby affecting the accuracy and validity of the structural model assessment. To address this concern, this study employed a full collinearity test, using the Variance Inflation Factor (VIF) to simultaneously detect both vertical collinearity (among predictor variables) and lateral collinearity (between predictor variables and criterion variables) [152], [153]. The VIF

TABLE 6. Discriminant validity - HTMT.

Variables	U/P	N	AA	VA	PE	ENJ	CF	PG	SC	CI
U/P										
N	.285									
AA	.319	.427								
VA	.362	.445	.606							
PE	.207	.337	.182	.167						
ENJ	.304	.310	.526	.647	.181					
CF	.284	.566	.463	.506	.293	.537				
PG	.145	.108	.328	.404	.229	.506	.260			
SC	.516	.299	.412	.581	.240	.632	.494	.343		
CI	.390	.427	.547	.718	.254	.813	.555	.415	.711	

U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.

TABLE 7. Collinearity assessment (Inner VIF values for structural model).

Variables	PE	ENJ	CF	PG	SC	CI
U/P	1.132	1.132	1.132	1.132	1.132	
N	1.238	1.238	1.238	1.238	1.238	
AA	1.465	1.465	1.465	1.465	1.465	
VA	1.511	1.511	1.511	1.511	1.511	
PE						1.067
ENJ						1.460
CF						1.313
PG						1.205
SC						1.250

U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.

value reflects the extent to which the variance of a particular predictor variable is inflated due to high correlations with other predictor variables [154]. The VIF values of the latent variables in this study are presented in TABLE 7.

According to the literature, VIF values should be below 2.4 to avoid collinearity issues that may interfere with the model [155]. Some studies also suggest that VIF values below 3.3 or 5 are considered acceptable [156]. In this study, all latent variables exhibited VIF values ranging from 1.067 to 1.511, which are well below the most stringent threshold of 2.4. This indicates that the structural model in this study is not affected by multicollinearity issues.

We analyzed the paths proposed in the structural model, with all path coefficients and significance levels presented in FIGURE 4 and TABLE 8.

We found that usability/playability did not significantly affect play engrossment ($\beta = 0.074$, $p = .128$), enjoyment ($\beta = 0.064$, $p = .148$), creative freedom ($\beta = 0.048$, $p = .223$), or personal gratification ($\beta = 0.016$, $p = .710$), indicating that hypotheses H1a, H1b, H1c, and H1d were not supported. However, usability/playability had a significant positive effect on social connectivity ($\beta = 0.218$, $p < .001$), supporting hypothesis H1e.

Additionally, narratives had a significant positive effect on play engrossment ($\beta = 0.212$, $p = < .001$) and creative freedom ($\beta = 0.335$, $p = < .001$), thus supporting hypotheses

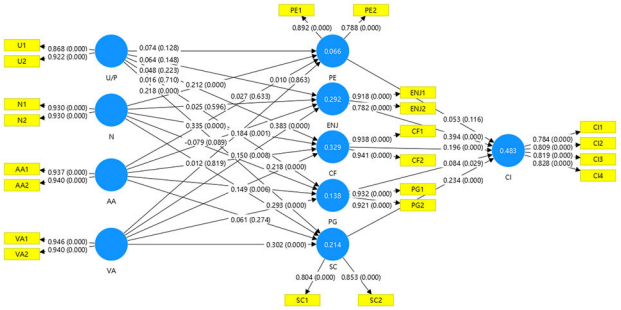


FIGURE 4. Structural model results.

H2a and H2c. However, narratives did not significantly affect enjoyment ($\beta = 0.025$, $p = .596$), personal gratification ($\beta = -0.079$, $p = .089$), or social connectivity ($\beta = 0.012$, $p = .819$), leading to the rejection of hypotheses H2b, H2d, and H2e.

We also observed that audio aesthetics did not significantly influence play engrossment ($\beta = 0.027$, $p = .633$) or social connectivity ($\beta = 0.061$, $p = .274$), indicating that H3a and H3e were not supported. However, audio aesthetics significantly affected enjoyment ($\beta = 0.184$, $p = .001$), creative freedom ($\beta = 0.150$, $p = .008$), and personal gratification ($\beta = 0.149$, $p = .006$), supporting hypotheses H3b, H3c, and H3d.

While visual aesthetics did not significantly influence play engrossment ($\beta = 0.010$, $p = .863$), indicating that H4a was not supported, it significantly impacted enjoyment ($\beta = 0.383$, $p < .001$), creative freedom ($\beta = 0.218$, $p < .001$), personal gratification ($\beta = 0.293$, $p < .001$), and social connectivity ($\beta = 0.302$, $p < .001$), supporting hypotheses H4b, H4c, H4d, and H4e.

Finally, we found that only play engrossment did not significantly affect continuance intention ($\beta = 0.053$, $p = .116$), meaning that H5 was not supported. However, enjoyment ($\beta = 0.394$, $p < .001$), creative freedom ($\beta = 0.196$, $p < .001$), personal gratification ($\beta = 0.084$, $p = .029$), and social connectivity ($\beta = 0.234$, $p < .001$) had significant effects on continuance intention, thus supporting hypotheses H6, H7, H8, and H9.

We conducted statistical analyses, as shown in TABLE 9, to assess the indirect effects of design factors on continuance intention through psychological factors.

We found that usability/playability did not significantly affect continuance intention through play engrossment ($\beta = 0.004$, $p = .342$), enjoyment ($\beta = 0.025$, $p = .158$), creative freedom ($\beta = 0.009$, $p = .240$), or personal gratification ($\beta = 0.001$, $p = .740$), indicating that H10a, H10b, H10c, and H10d were not supported. However, usability/playability had a significant effect on continuance intention through social connectivity ($\beta = 0.051$, $p < .001$), supporting H10e.

Additionally, narratives did not significantly affect continuance intention through play engrossment ($\beta = 0.011$, $p = .153$) or enjoyment ($\beta = 0.010$, $p = .600$). Thus, H11a and H11b were not supported. However, narratives had a significant effect on continuance intention through creative freedom ($\beta = 0.065$, $p < .001$), supporting H11c. Narratives did not significantly influence continuance intention through personal gratification ($\beta = -0.007$, $p = .194$) or social connectivity ($\beta = 0.003$, $p = .821$), leading to the rejection of H11d and H11e.

TABLE 8. Structural model outcomes.

Hypothesis	β	Mean	STDEV	t-value	p-value	Supported
H1a U/P -> PE	0.074	0.076	0.048	1.522	.128	No
H1b U/P -> ENJ	0.064	0.064	0.044	1.445	.148	No
H1c U/P -> CF	0.048	0.049	0.039	1.218	.223	No
H1d U/P -> PG	0.016	0.017	0.042	0.372	.710	No
H1e U/P -> SC	0.218	0.221	0.043	5.125	<.001	Yes
H2a N -> PE	0.212	0.214	0.050	4.259	<.001	Yes
H2b N -> ENJ	0.025	0.025	0.048	0.530	.596	No
H2c N -> CF	0.335	0.334	0.051	6.602	<.001	Yes
H2d N -> PG	-0.079	-0.079	0.046	1.702	.089	No
H2e N -> SC	0.012	0.014	0.051	0.229	.819	No
H3a AA -> PE	0.027	0.027	0.058	0.477	.633	No
H3b AA -> ENJ	0.184	0.182	0.054	3.432	.001	Yes
H3c AA -> CF	0.150	0.149	0.056	2.669	.008	Yes
H3d AA -> PG	0.149	0.149	0.054	2.750	.006	Yes
H3e AA -> SC	0.061	0.063	0.056	1.095	.274	No
H4a VA -> PE	0.010	0.010	0.060	0.173	.863	No
H4b VA -> ENJ	0.383	0.387	0.057	6.733	<.001	Yes
H4c VA -> CF	0.218	0.220	0.053	4.132	<.001	Yes
H4d VA -> PG	0.293	0.294	0.053	5.546	<.001	Yes
H4e VA -> SC	0.302	0.299	0.057	5.276	<.001	Yes
H5 PE -> CI	0.053	0.054	0.033	1.573	.116	No
H6 ENJ -> CI	0.394	0.395	0.045	8.749	<.001	Yes
H7 CF -> CI	0.196	0.196	0.037	5.263	<.001	Yes
H8 PG -> CI	0.084	0.083	0.038	2.187	.029	Yes
H9 SC -> CI	0.234	0.234	0.042	5.608	<.001	Yes

H = hypothesis, U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.

TABLE 9. Mediation affects outcomes.

Hypothesis	β	Mean	STDEV	t-value	p-value	Supported
H10a U/P -> PE -> CI	0.004	0.004	0.004	0.950	.342	No
H10b U/P -> ENJ -> CI	0.025	0.025	0.018	1.413	.158	No
H10c U/P -> CF -> CI	0.009	0.009	0.008	1.175	.240	No
H10d U/P -> PG -> CI	0.001	0.001	0.004	0.332	.740	No
H10e U/P -> SC -> CI	0.051	0.052	0.014	3.687	<.001	Yes
H11a N -> PE -> CI	0.011	0.012	0.008	1.428	.153	No
H11b N -> ENJ -> CI	0.010	0.010	0.019	0.524	.600	No
H11c N -> CF -> CI	0.065	0.066	0.017	3.925	<.001	Yes
H11d N -> PG -> CI	-0.007	-0.006	0.005	1.300	.194	No
H11e N -> SC -> CI	0.003	0.003	0.012	0.227	.821	No
H12a AA -> PE -> CI	0.001	0.001	0.004	0.388	.698	No
H12b AA -> ENJ -> CI	0.072	0.071	0.022	3.344	.001	Yes
H12c AA -> CF -> CI	0.029	0.029	0.013	2.269	.023	Yes
H12d AA -> PG -> CI	0.012	0.013	0.008	1.537	.124	No
H12e AA -> SC -> CI	0.014	0.015	0.013	1.073	.283	No
H13a VA -> PE -> CI	0.001	0.001	0.004	0.139	.890	No
H13b VA -> ENJ -> CI	0.151	0.153	0.031	4.838	<.001	Yes
H13c VA -> CF -> CI	0.043	0.043	0.014	3.131	.002	Yes
H13d VA -> PG -> CI	0.025	0.025	0.012	1.967	.049	Yes
H13e VA -> SC -> CI	0.071	0.071	0.020	3.455	.001	Yes

H = hypothesis, U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.

TABLE 10. Explanatory and predictive power of the structural model.

Variables	R^2	R^2 Adjusted	Q^2
PE	0.066	0.059	0.045
ENJ	0.292	0.287	0.275
CF	0.329	0.324	0.315
PG	0.138	0.132	0.123
SC	0.214	0.207	0.197
CI	0.483	0.478	0.378

PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.

We also observed that audio aesthetics did not significantly affect continuance intention through play engrossment (β = 0.001, p = .698) or social connectivity (β = 0.014, p = .283), indicating that H12a and H12e were not supported. However, audio aesthetics significantly affected continuance intention through enjoyment (β = 0.072, p = .001) and creative freedom (β = 0.029, p = .023), thus supporting H12b and H12c. Audio aesthetics did not significantly affect continuance intention through personal gratification (β = 0.012, p = .124), indicating that H12d was not supported.

In addition to the above findings, visual aesthetics did not significantly influence continuance intention through play engrossment (β = 0.001, p = .892), meaning that H13a was not supported. However, visual aesthetics had a significant effect on continuance intention through enjoyment (β = 0.151, p< .001), creative freedom (β = 0.043, p = .002), personal gratification (β = 0.025, p = .049), and social connectivity (β = 0.071, p = .001), thus supporting H13b, H13c, H13d, and H13e.

This study further evaluated the structural model's explanatory and predictive capabilities of the structural model through R^2 (R-square), Adjusted R^2, and Predictive Relevance (Q^2). The specific results are presented in TABLE 10.

The R^2 value (Coefficient of Determination) is a crucial indicator for assessing the extent to which endogenous variables (dependent variables) are explained by predictor variables (independent variables) within a structural model. Its value ranges from 0 to 1, with higher values indicating more substantial explanatory power [157]. The Adjusted R^2 further refines the R^2 value by accounting for the number of predictor variables and sample size, thereby mitigating the potential overestimation of explanatory power. This adjustment provides a more conservative and accurate evaluation of the model's explanatory ability, mainly when the sample size is relatively small or the model includes numerous independent variables. Notably, the Adjusted R^2 is often lower than the R^2 and may even be negative when the model's explanatory power is inadequate, indicating a need for careful interpretation [158]. According to recommended standards, R^2 values of 0.75, 0.50, and 0.25 indicate strong, moderate, and weak explanatory power, respectively [159].

In this study, the R^2 values of the structural model range from 0.066 to 0.483, while the Adjusted R^2 values range from 0.059 to 0.478. Specifically, continuance intention (R^2 = 0.483; Adjusted R^2 = 0.478) approaches a moderate level of explanatory power. creative freedom (R^2 = 0.329; Adjusted R^2 = 0.324), enjoyment (R^2 = 0.292; Adjusted R^2 = 0.287), and social connectivity (R^2 = 0.214; Adjusted R^2 = 0.207) demonstrate weak explanatory power. Additionally, personal gratification (R^2 = 0.138; Adjusted R^2 = 0.132) and play engrossment (R^2 = 0.066; Adjusted R^2 = 0.059) exhibit relatively low explanatory power.

TABLE 11. Model fitness.

Variables	Saturated Model	Estimated Model
SRMR	0.070	0.089

Furthermore, this study applied Stone-Geisser's Q^2 value to evaluate the structural model's predictive relevance [160], [161]. A Q^2 value greater than 0 indicates that the model has sufficient predictive capability for the endogenous constructs [162]. All Q^2 values for the endogenous constructs in this study are greater than 0, ranging from 0.045 to 0.378, indicating that the structural model possesses adequate predictive relevance.

Specifically, the Q^2 value for continuance intention is the highest (CI, Q^2 = 0.378), followed by creative freedom (CF, Q^2 = 0.315) and enjoyment (ENJ, Q^2 = 0.275), which demonstrate moderate predictive relevance. In contrast, social connectivity (SC, Q^2 = 0.197), personal gratification (PG, Q^2 = 0.123), and play engrossment (PE, Q^2 = 0.045) exhibit relatively low predictive relevance.

Finally, this study employed the Standardized Root Mean Square Residual (SRMR) to evaluate the model fit of the structural model [163]. The specific results are presented in TABLE 11.

This study employed the SRMR to evaluate the structural model's fit [164], although in practical research, an SRMR value below 0.10 is also considered acceptable [165]. In this study, the SRMR value for the Saturated Model was 0.070, indicating good model fit. At the same time, the Estimated Model had an SRMR value of 0.089, which falls within the acceptable fit range.

D. MULTIGROUP INVARIANCE

This study used PLS-MGA to investigate differences in the path model between casual players (n = 115) and core players (n = 219). Bootstrapping was conducted with a significance level of 0.05 and 5,000 sample iterations to facilitate random sampling by the system for calculating the results, as shown in TABLE 12 and TABLE 13.

The PLS-MGA analysis results indicated that the two groups showed significant differences in the structural model's direct effect paths only for H3b. In contrast, significant differences in the indirect effect paths appeared only for H11d.

Regarding direct effects, there was a significant difference in the impact of audio aesthetics on enjoyment, with casual

TABLE 12. Multigroup invariance results of mediation effects between Casual player and Core Player groups.

Hypothesis		β_{casual}	β_{core}	$\Delta\beta$	p-value
H1a	U/P -> PE	0.112	0.029	0.083	.552
H1b	U/P -> ENJ	0.048	0.054	-0.006	.961
H1c	U/P -> CF	0.063	0.099	-0.036	.701
H1d	U/P -> PG	-0.045	-0.016	-0.028	.804
H1e	U/P -> SC	0.224**	0.195**	0.029	.780
H2a	N -> PE	0.365***	0.129	0.236	.090
H2b	N -> ENJ	0.008	-0.061	0.069	.549
H2c	N -> CF	0.347***	0.322***	0.025	.828
H2d	N -> PG	-0.005	-0.097	0.092	.441
H2e	N -> SC	-0.043	-0.018	-0.025	.841
H3a	AA -> PE	0.093	0.034	0.059	.694
H3b	AA -> ENJ	0.403***	0.109	0.294*	.031
H3c	AA -> CF	0.197	0.16	0.037	.768
H3d	AA -> PG	0.193	0.166*	0.027	.833
H3e	AA -> SC	0.107	0.098	0.008	.968
H4a	VA -> PE	-0.003	0.038	-0.041	.797
H4b	VA -> ENJ	0.208	0.475***	-0.267	.051
H4c	VA -> CF	0.27**	0.178*	0.092	.469
H4d	VA -> PG	0.17	0.225**	-0.055	.673
H4e	VA -> SC	0.311*	0.213**	0.097	.543
H5	PE -> CI	0.06	0.072	-0.011	.921
H6	ENJ -> CI	0.294**	0.459***	-0.164	.148
H7	CF -> CI	0.14	0.209***	-0.069	.490
H8	PG -> CI	0.102	0.001	0.101	.254
H9	SC -> CI	0.314**	0.204***	0.110	.374

H = hypothesis, U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.
$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$.

TABLE 13. Multigroup invariance results of structural models between casual player and core player groups.

Hypothesis		β_{casual}	β_{core}	$\Delta\beta$	p-value
H10a	U/P -> PE -> CI	0.007	0.002	0.005	.747
H10b	U/P -> ENJ -> CI	0.014	0.025	-0.011	.775
H10c	U/P -> CF -> CI	0.009	0.021	-0.012	.488
H10d	U/P -> PG -> CI	-0.005	-1.425×10^{-5}	-0.005	.648
H10e	U/P -> SC -> CI	0.07*	0.04*	0.031	.447
H11a	N -> PE -> CI	0.022	0.009	0.013	.656
H11b	N -> ENJ -> CI	0.002	-0.028	0.03	.491
H11c	N -> CF -> CI	0.049	0.067**	-0.019	.632
H11d	N -> PG -> CI	-4.781×10^{-4}	-8.4×10^{-5}	-3.942×10^{-4}	.986
H11e	N -> SC -> CI	-0.013	-0.004	-0.01	.808
H12a	AA -> PE -> CI	0.006	0.002	0.003	.869
H12b	AA -> ENJ -> CI	0.119***	0.05	0.069	.150
H12c	AA -> CF -> CI	0.028	0.033	-0.006	.829
H12d	AA -> PG -> CI	0.02	1.443×10^{-4}	0.02	.360
H12e	AA -> SC -> CI	0.033	0.02	0.013	.788
H13a	VA -> PE -> CI	-2.048×10^{-4}	0.003	-0.003	.721
H13b	VA -> ENJ -> CI	0.061	0.218***	-0.157*	.032
H13c	VA -> CF -> CI	0.038	0.037	0.001	.995
H13d	VA -> PG -> CI	0.017	1.949x10-4	0.017	.431
H13e	VA -> SC -> CI	0.098	0.043	0.054	.478

H = hypothesis, U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.
$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$.

players showing a more substantial effect than core players ($\beta_{casual} = 0.403$, $\beta_{core=}0.109$, $\Delta\beta = 0.294$, $p= .031$).

For indirect effects, a significant difference was found in the impact of visual aesthetics on continuance intention through the mediation of enjoyment, with core players showing a more substantial effect than casual players ($\beta_{casual} = 0.061$, $\beta_{core} = 0.218$, $\Delta\beta = -0.157$, $p= .032$).

We conducted tests using PLS-MGA to understand the differences in the path model for *League of Legends* on desktop platforms (n = 413) and mobile platforms (n = 111). The results are presented in TABLE 14 and TABLE 15.

V. DISCUSSION

Although many studies have explored game experience satisfaction from different perspectives, extending the investigation to its impact on continuance intention, there is still a lack of academic research that delves deeply into the potential mediating effects among these factors. Based on previous research theories that examine game design, game systems, psychological characteristics, and behavioral changes, this study posits that mediating processes exist between these factors, which serves as the main focus of this research [28].

A. DEMOGRAPHIC CHARACTERISTICS OF PARTICIPANTS

The results of this study indicate that desktop and mobile players share similar demographic characteristics in most aspects. However, significant differences were found in terms of educational level and gaming experience.

Specifically, a higher proportion of mobile players have lower educational levels (high school or below) compared to desktop players. Previous studies have suggested that individuals with lower educational attainment are more likely to use mobile games due to the accessibility of mobile devices, while those with higher educational levels may prefer using desktops for gaming due to possessing better digital skills and equipment and a preference for more diverse internet usage [166].

The PLS-MGA analysis indicated that there were no significant differences between the two groups across all direct and indirect effect paths in the structural model for different device platforms. Thus, none of the hypothesized paths varied based on the platform used.

Additionally, mobile players generally have a shorter gaming experience, mostly ranging from 2 to 4 years, whereas desktop players are predominantly long-term gamers with over 4 years of experience. This difference may reflect the fact that *League of Legends: Wild Rift*, released in 2020 as a mobile platform game, is relatively new compared to the desktop-based *League of Legends*, which was launched in 2009. This observation also supports the finding that mobile players tend to have shorter gaming experiences [8], [13].

However, it is worth noting that although there are significant differences in gaming experience between the two groups, there is no significant difference in weekly gaming hours. This suggests that the level of engagement in gameplay is similar for both groups, indicating that even though mobile players have less overall gaming experience, the amount of time they invest in gaming is comparable to that of desktop players.

TABLE 14. Multigroup invariance results of structural models between desktop and mobile device groups.

Hypothesis		$\beta_{desktop}$	β_{mobile}	$\Delta\beta$	p-value
H1a	U/P -> PE	0.071	0.116	-0.045	.683
H1b	U/P -> ENJ	0.081	-0.011	0.092	.342
H1c	U/P -> CF	0.083	-0.072	0.155	.151
H1d	U/P -> PG	0.008	0.034	-0.026	.804
H1e	U/P -> SC	0.199***	0.308**	-0.109	.302
H2a	N -> PE	0.231***	0.093	0.138	.400
H2b	N -> ENJ	-0.014	0.18	-0.194	.084
H2c	N -> CF	0.343***	0.313**	0.030	.821
H2d	N -> PG	-0.073	-0.109	0.036	.718
H2e	N -> SC	$-4.28\text{x}10^{-5}$	0.028	-0.028	.817
H3a	AA -> PE	0.017	0.121	-0.104	.545
H3b	AA -> ENJ	0.185**	0.205*	-0.020	.866
H3c	AA -> CF	0.148*	0.171	-0.023	.885
H3d	AA -> PG	0.155**	0.078	0.077	.646
H3e	AA -> SC	0.084	-0.075	0.160	.270
H4a	VA -> PE	0.039	-0.143	0.181	.297
H4b	VA -> ENJ	0.398***	0.322**	0.076	.566
H4c	VA -> CF	0.227***	0.169	0.058	.666
H4d	VA -> PG	0.26***	0.457***	-0.197	.175
H4e	VA -> SC	0.291***	0.445***	-0.154	.281
H5	CF -> CI	0.059	0.034	0.025	.719
H6	ENJ -> CI	0.39***	0.429***	-0.040	.782
H7	PE -> CI	0.213***	0.105	0.108	.208
H8	PG -> CI	0.061	0.137	-0.076	.589
H9	SC -> CI	0.224***	0.291***	-0.066	.479

H = hypothesis, U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.
*p<0.05, **p<0.01, ***p<0.001.

TABLE 15. Multigroup invariance results of mediation effects between desktop and mobile device groups.

Hypothesis		$\beta_{desktop}$	β_{mobile}	$\Delta\beta$	p-value
H10a	U/P -> PE -> CI	0.004	0.004	$2.617\text{x}10^{-4}$	.879
H10b	N -> PE -> CI	0.032	$-4.661\text{x}10^{-3}$	0.036	.373
H10c	AA -> PE -> CI	0.018	$-7.572\text{x}10^{-3}$	0.025	.085
H10d	VA -> PE -> CI	$4.838\text{x}10^{-4}$	0.005	-0.004	.663
H10e	U/P -> ENJ -> CI	0.045**	0.089*	-0.045	.307
H11a	N -> ENJ -> CI	0.014	0.003	0.011	.447
H11b	AA -> ENJ -> CI	$-5.349\text{x}10^{-3}$	0.077	-0.083	.096
H11c	VA -> ENJ -> CI	0.073***	0.033	0.04	.255
H11d	U/P -> CF -> CI	$-4.427\text{x}10^{-3}$	$-1.484\text{x}10^{-2}$	0.01	.637
H11e	N -> CF -> CI	$-9.597\text{x}10^{-6}$	0.008	-0.008	.821
H12a	AA -> CF -> CI	0.001	0.004	-0.003	.989
H12b	VA -> CF -> CI	0.072**	0.088	-0.016	.858
H12c	U/P -> PG -> CI	0.032*	0.018	0.014	.534
H12d	N -> PG -> CI	0.009	0.011	-0.001	.763
H12e	AA -> PG -> CI	0.019	$-2.19\text{x}10^{-2}$	0.041	.302
H13a	VA -> PG -> CI	0.002	$-4.889\text{x}10^{-3}$	0.007	.680
H13b	U/P -> SC -> CI	0.155***	0.138*	0.017	.775
H13c	N -> SC -> CI	0.048**	0.018	0.031	.228
H13d	AA -> SC -> CI	0.016	0.062	-0.047	.451
H13e	VA -> SC -> CI	0.065**	0.129**	-0.064	.239

H = hypothesis, U/P = usability/playability, N = narratives, AA = audio aesthetics, VA = visual aesthetics, PE = play engrossment, ENJ = enjoyment, CF = creative freedom, PG = personal gratification, SC = social connectivity, and CI = continuance intention.
*p<0.05, **p<0.01, ***p<0.001.

B. DIRECT EFFECTS

Our findings reveal that usability/playability does not significantly influence play engrossment, enjoyment, creative freedom, or personal gratification, but it does have a significant positive effect on social connectivity. This result aligns with previous studies indicating that the usability/playability of video games is a key driver of players' social interactions [19], [52]. This suggests that in eSports games, usability/playability mainly impacts the interactions between players.

Narratives significantly influence play engrossment and creative freedom but do not significantly affect enjoyment, personal gratification, or social connectivity. This indicates that narrative features in games can enhance players' immersion and flow experience [55], [56]. By constructing worldviews through game narratives, players connect with the game world, engage in critical thinking about specific scenarios, and strengthen their creativity and imagination [61], [62]. In particular, the recent animated series *Arcane*, based on *League of Legends*, has expanded the narrative universe significantly, potentially enhancing players' emotional connections to characters and storylines, thereby deepening their immersion and engagement with the game's narrative elements [167]. However, the limited effects in other areas suggest that narratives primarily enhance the depth and complexity of the game's worldview.

Audio aesthetics significantly affect enjoyment, creative freedom, and personal gratification but do not significantly impact play engrossment or social connectivity. Synchronized game sound effects are known to bring pleasure [168], and sound is an essential tool in interactive environments that affects players' emotions and imagination [80], [81]. In particular, music in games has been shown to play a crucial role in players' self-challenges [83]. However, its influence on immersion and social connectivity is more minor, suggesting that further user-centered research is needed to better understand the causal relationship between audio aesthetics [75].

Visual aesthetics significantly affect enjoyment, creative freedom, personal gratification, and social connectivity but do not have a significant effect on play engrossment. Attractive visual design and rich visual feedback can enhance players' pleasurable gaming experiences [95], [96]. The richness and diversity of visual aesthetics stimulate players' imagination, thereby increasing creative freedom and providing unlimited inspiration [98], [99]. Engaging visual elements triggers positive emotions and enhances users' sense of self-worth [100], [101]. Friendly visual aesthetic features and atmosphere can increase users' willingness to interact and strengthen social connections [103], [104]. However, visual elements may affect players' aesthetic and overall experiences more than directly enhancing immersion in the game world.

We also found that play engrossment does not significantly influence continuance intention, while enjoyment, creative freedom, personal gratification, and social connectivity do have significant effects on it. Enhancing pleasure has been shown in previous research to promote continuance intention effectively [46], [109], [110]. Imagination in the virtual world has been proven to strengthen players' emotional attachment

to the game and increase continuance intention [66], [111]. Players not only experience pleasure from the game but also develop a sense of identity and achievement through gameplay, leading to satisfaction and continuance intention [114], [115]. As online games are socially interactive, players who establish strong social relationships through social features show higher loyalty to continue playing [19], [118].

C. MEDIATION EFFECTS

Additionally, we conducted statistical analyses to assess the indirect effects of design factors on continuance intention through psychological factors. The results show that usability/playability significantly influences continuance intention through social connectivity but not through other psychological factors. The interaction between usability/playability and social connectivity has also been confirmed as a key mediating factor affecting the word-of-mouth promotion of multiplayer online games [47].

Narratives significantly influence continuance intention through creative freedom, but not through other psychological factors. This finding diverges somewhat from previous studies, possibly because the narratives in eSports games primarily focus on providing players with a fantastical worldview [122].

Audio aesthetics significantly influence continuance intention through enjoyment and creative freedom, but not through other psychological factors. Similar findings have been confirmed in research exploring perceived auditory feedback in mobile games, although creative freedom was not explicitly investigated [78].

Visual aesthetics significantly influence continuance intention through enjoyment, creative freedom, personal gratification, and social connectivity, but not through play engrossment. Visual aesthetics have also been proven to be essential influencers of user attitudes toward games, encouraging users to continue playing, meeting their psychological needs, and affecting both word-of-mouth and continuance intention [46], [123].

D. MULTIGROUP INVARIANCE

Through PLS-MGA analysis, we found that the differences between casual and core players in the structural model mainly lie in the effects of audio aesthetics on enjoyment. Specifically, the influence of audio aesthetics on enjoyment was significantly higher for casual players than for core players. Casual players are more likely to be driven by immediate emotional feedback, making audio aesthetics more impactful in providing instant enjoyment during gameplay [14]. In contrast, core players are more substantially indirectly affected by visual aesthetics when it comes to enhancing enjoyment and promoting continuance intention. These differences in overall game user experience may be attributed to factors such as gameplay time, skill level, and attitudes toward the game [35], [36]. Core players may have higher expectations for game quality and depth, making visual aesthetics more effective in enhancing their enjoyment and reinforcing their intention to continue playing. In contrast, casual players are more likely to be driven by immediate emotional feedback, making audio aesthetics more impactful in providing instant enjoyment during gameplay [14]. Notably, this study not only confirms these differences but also reveals that visual aesthetics significantly mediate the effect of enjoyment on continuance intention, with core players showing a more substantial effect than casual players.

We also used PLS-MGA to examine the impact of different gaming platforms (desktop and mobile) on the path model. The results showed no significant differences between players on different platforms for all direct and indirect paths, indicating that the gaming platform does not significantly influence continuance intention. Previous research has confirmed that users experience significant differences in satisfaction with graphics and controls when playing video games on desktop computers versus smartphones [29], [30]. However, this study yielded non-significant results, which may be due to the fact that prior studies often compared games with substantial differences in content and controls. In contrast, this study focused on games from the same company and brand, potentially reducing the differences in user experience across platforms.

E. LIMITATIONS

This study has several limitations. First, it focuses on MOBA games and only includes the desktop and mobile platform of *League of Legends* for the investigation. While popular eSports games today also include first-person shooter and real-time strategy genres, there may be differences in sustained attention, reaction time, strategy, and cognition between these different game types [169], [170].

The second limitation is that the sample primarily consists of participants aged 20-30, with the majority being male. Previous studies have found that young males tend to prefer strategy, sports, simulation, and action games, while older and female players tend to prefer casual games [171]. In research on MOBA games, motivational theories used to explore factors influencing user loyalty have shown that males exhibit significantly higher loyalty than females [108]. Although this phenomenon has been observed in other studies, it inevitably raises concerns about the generalizability of the findings.

Third, the internal consistency of certain constructs within the GUESS framework was relatively low in this study. Specifically, the internal consistency for Enjoyment ranged between 0.6 and 0.7, while Play Engrossment and Social Connectivity ranged between 0.5 and 0.6, which may affect the interpretation of the results. Previous research on measurement length and reliability has indicated that shorter scales can result in lower reliability estimates, mainly when only two items are included. In such cases, Cronbach's Alpha is directly equivalent to the correlation between items, potentially underestimating the true reliability [172].

Moreover, prior studies utilizing the GUESS-18 have also reported slightly lower Alpha values for certain constructs, falling below 0.7 [173]. To address this issue, some studies recommend supplementing reliability assessments with the Spearman-Brown coefficient (SB), which is particularly suitable for evaluating the internal consistency of two-item scales and can provide more reliable reliability estimates [174]. Comparative studies between GUESS-55 and GUESS-18 have revealed that six out of the nine constructs in GUESS-18 demonstrated Spearman-Brown coefficients ranging from 0.5 to 0.7, indicating a potential reliability bias due to shorter scales [175]. Low reliability may increase measurement error and affect the accuracy of the research findings, highlighting the need for cautious interpretation and discussion of the results. Future research could consider increasing the number of items or revising the design of the Game User Experience Scales to enhance the stability and reliability of the scales [176], [177].

F. THEORETICAL IMPLICATIONS

This research critically examines the impact of game user experience satisfaction on continuance intention by examining game design, activity satisfaction, and behavioral engagement, proposing a framework that distinguishes between design factors and psychological factors in influencing continuance intention [27]. This framework aligns with the foundational theories of TAM. However, previous research has predominantly focused on the overall GUESS score's influence on continuance intention [25]. In contrast, this study further divides game user experience into design factors and psychological factors. It verifies the mediating effects among these constructs, thereby addressing the gaps in the literature concerning the relationship between game user experience and continuance intention. Specifically, this study presents several critical theoretical findings.

First, enhancing communication and interaction features within games can significantly strengthen the positive influence of usability/playability on social connectivity. This finding supports the notion that social relatedness plays a crucial role in influencing user satisfaction, a relationship that has been relatively underexplored in previous research [19]. Moreover, expanding narratives through transmedia approaches, such as the animated series Arcane, effectively enhances the impact of narratives on creative freedom, thereby increasing users' immersion and emotional attachment [167]. This demonstrates that narratives in eSports games are not merely background settings but essential elements that promote creativity and player engagement. Additionally, audio and visual aesthetics have significant effects on enhancing enjoyment, personal gratification, and creative freedom [55], [86], [94]. These insights underscore the pivotal role of aesthetic experiences in influencing player satisfaction and continuance intention, supporting previous theories suggesting that visual and auditory designs are critical in shaping players' psychological states.

These findings not only confirm the mediating effects of different constructs within user experience but also expand the current theoretical understanding of players' behavioral intentions. Future research could further examine the stability and effectiveness of these mediating effects and explore the differences between various game types, platforms, and player groups. Such detailed analysis will contribute to constructing more accurate theoretical models and provide practical recommendations for game design and user experience optimization.

VI. CONCLUSION

This study confirms that design and psychological factors in game user experience satisfaction significantly influence players' intention to continue playing the game. Enjoyment, creative freedom, personal gratification, and social connectivity all have a substantial impact on players' willingness to continue engaging with the game.

The findings indicate that although desktop and mobile players are similar across most demographic characteristics, significant differences exist in education levels and gaming experience. Mobile players are more likely to have lower educational attainment and shorter gaming experience, while desktop players tend to have higher education levels and longer gaming histories. However, the level of gaming involvement between the two groups is not significantly different, suggesting that despite having shorter gaming experience, mobile players spend similar amounts of time playing games as desktop players.

Additionally, visual aesthetics demonstrated the most robust and comprehensive mediating effects on continuance intention through enjoyment, creative freedom, personal gratification, and social connectivity, highlighting the central role of visual design in enhancing players' long-term engagement motivation. This study also reveals that audio aesthetics not only significantly influence personal gratification, enjoyment, and creative freedom, but also indirectly affect continuance intention through enjoyment and creative freedom.

On the other hand, while usability and playability do not directly influence most psychological factors, they indirectly enhance players' continuance intention by strengthening social connectivity, underscoring the importance of interactive design in promoting social engagement. Furthermore, game narratives influence play engrossment and creative freedom, indirectly affecting continuance intention through creative freedom, indicating the importance of deep and creative storytelling in enhancing immersion and even creative thinking.

It is worth noting that through PLS-MGA, this study further reveals differences in game user experience satisfaction between core and casual players: audio aesthetics have a more pronounced impact on enjoyment for casual players. In contrast, visual aesthetics have a more substantial indirect

influence on enjoyment for core players, which in turn further reinforces their continuance intention. Although platform differences between desktop and mobile were not found to have significant effects, this may be due to the high similarity of game content and features across both platforms in the selected game, thereby reducing the impact of platform differences.

Theoretically, this study enriches the understanding of game design and user experience satisfaction by identifying the complex mediating relationships between design and psychological factors and their connection to continuance intention. Future research could further explore different game genres and develop more reliable, valid, and concise game user experience scales to understand better the design and psychological factors that drive players' sustained engagement. Additionally, game developers should focus on enhancing visual and audio quality, strengthening narrative structures, improving interaction design, and enhancing social mechanisms to boost user experience satisfaction and long-term loyalty comprehensively.

DATA AVAILABILITY STATEMENT

The data supporting this study are not publicly available due to ethical review requirements and data confidentiality regulations.

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